

5.1 Representation of data

Probability and Statistics 1 · 2026 examination · Concept guide and original exam-style practice

Understand the concept

Choose displays suited to data type; histogram area, not height, represents frequency if widths differ. Cumulative frequency estimates median and percentiles; mean of grouped data uses midpoints and is approximate. Standard deviation measures spread about the mean, unlike range which relies on extremes.

Worked example: follow the reasoning

A class width 5 with frequency 20 has frequency density 4; a width 10 class with frequency 20 has height 2.

Structured practice

Answer in the space provided. Show working and units where relevant.

- 1 Two histogram classes have frequencies 12 and 18 with widths 3 and 6. Find both frequency densities. [2]

Final answer: _____

Answers and explanation

Question 1 [2 marks]

$12/3=4$ [1]; $18/6=3$ [1].

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or calculation error; then retry without looking at the solution.

Syllabus: Mathematics 9709, 2026–2027. <https://www.cambridgeinternational.org/Images/697427-2026-2027-syllabus.pdf>

Original FagusED teaching material. Question layout and mark presentation informed by the May/June 2026 papers supplied by the user. The practice question is newly written and is not a Cambridge past-paper question.